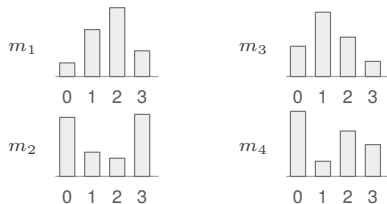


(a) Native sequence-dependent kernel

Motif identity m and segment s define

$$P_0(z \mid m, s)$$

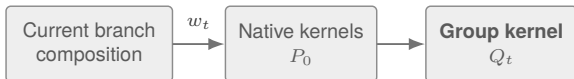


Symbolic post-softmax probability profiles over local-state class $z \in \{0, 1, 2, 3\}$.

(c) Probability-space aggregation

Within each group q and segment s , current intervention-branch frequencies $w_t(m \mid q, s)$ weight the native kernels:

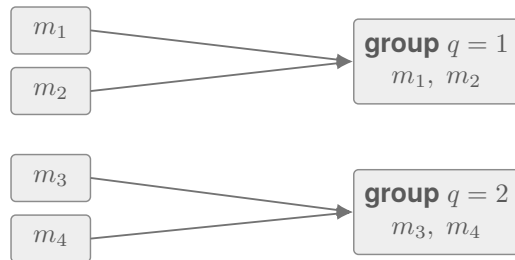
$$Q_t(z \mid q, s) = \sum_{m: g(m)=q} w_t(m \mid q, s) P_0(z \mid m, s)$$



Weights are recomputed from the current intervention-branch population, not fixed from the baseline population.

(b) Grouping map

$g(m) = q$ assigns motif identities to grouping classes.



Sequence identities are unchanged; grouping controls which motif distinctions are retained.

(d) Intervention and preservation identity

All motifs in group q
use the same Q_t

Removed

within-group motif-specific local-state dependence

Preserved

expected one-site $P(Z \mid S)$

Not forced to remain fixed

adjacency-category frequencies and downstream model fitness

$$\sum_q P_t(q \mid s) Q_t(z \mid q, s) = \sum_m P_t(m \mid s) P_0(z \mid m, s)$$